

M65 - Streak vs switch-rate correlation and effect

PER-QUESTION CORRELATIONS (each row = pair across all clean questions)

Pair	n	Pearson r	Pear p	Spears r
seq_typing_mean_run_explore vs n_switches_typing	165	-0.293	0.000136	-0.296
seq_typing_mean_run_explore vs switch_rate_typing	165	-0.666	1.65e-22	-0.834
seq_typing_mean_run_explore vs switch_rate_topic	165	+0.019	0.809	+0.012
seq_typing_mean_run_explore vs switch_rate_time	165	-0.233	0.00259	-0.230
seq_typing_entropy vs n_switches_typing	172	+0.393	9.71e-08	+0.262
seq_typing_entropy vs switch_rate_typing	172	+0.670	9.07e-24	+0.611
n_switches_typing vs switch_rate_typing	172	+0.627	3.53e-20	+0.666

PER-PARTICIPANT CORRELATIONS (averaged across both questions)

Pair	n	Pearson r	Pear p	Spears r
seq_typing_mean_run_explore vs n_switches_typing	99	-0.191	0.0585	-0.196
seq_typing_mean_run_explore vs switch_rate_typing	99	-0.638	1.19e-12	-0.725
seq_typing_mean_run_explore vs switch_rate_topic	99	+0.155	0.125	+0.165
seq_typing_mean_run_explore vs switch_rate_time	99	-0.184	0.0678	-0.164
seq_typing_entropy vs n_switches_typing	101	+0.411	2.01e-05	+0.312
seq_typing_entropy vs switch_rate_typing	101	+0.645	3.18e-13	+0.597
n_switches_typing vs switch_rate_typing	101	+0.576	2.84e-10	+0.650

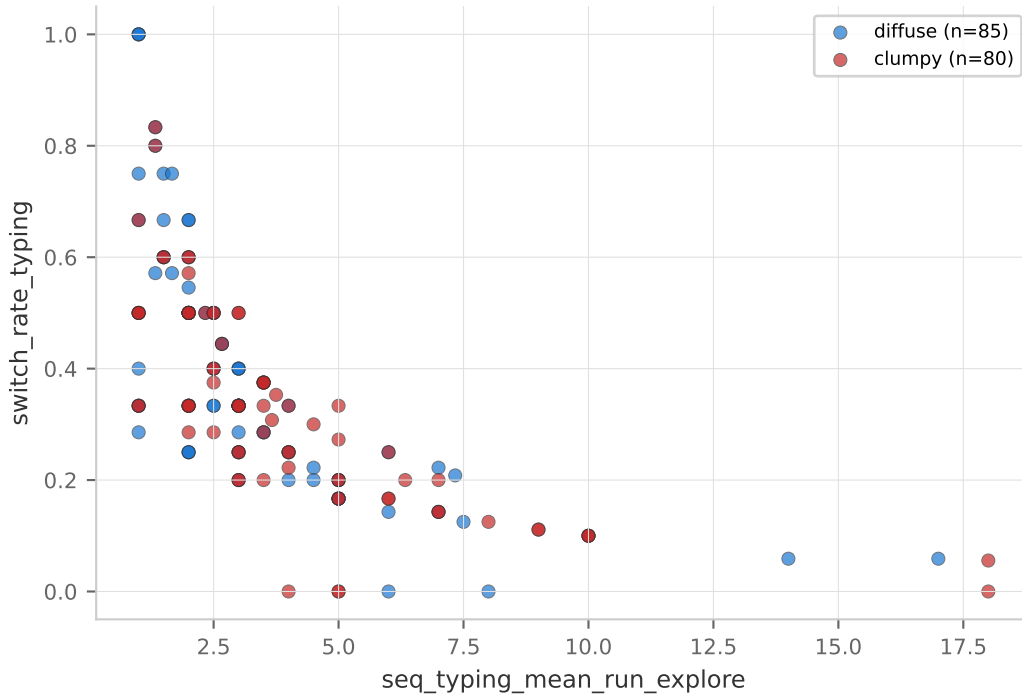
EFFECT (Diffuse - Clumpy), per-participant averaged values

Measure	d_mean	c_mean	raw d	raw p	adj d	adj p
n_switches_typing	2.019	1.847	+0.16	0.435	+0.29	0.160
switch_rate_typing	0.383	0.305	+0.48	0.017	+0.57	0.007
seq_typing_mean_run_explore	3.258	3.925	-0.32	0.122	-0.44	0.038
seq_typing_entropy	0.782	0.669	+0.52	0.012	+0.69	0.001

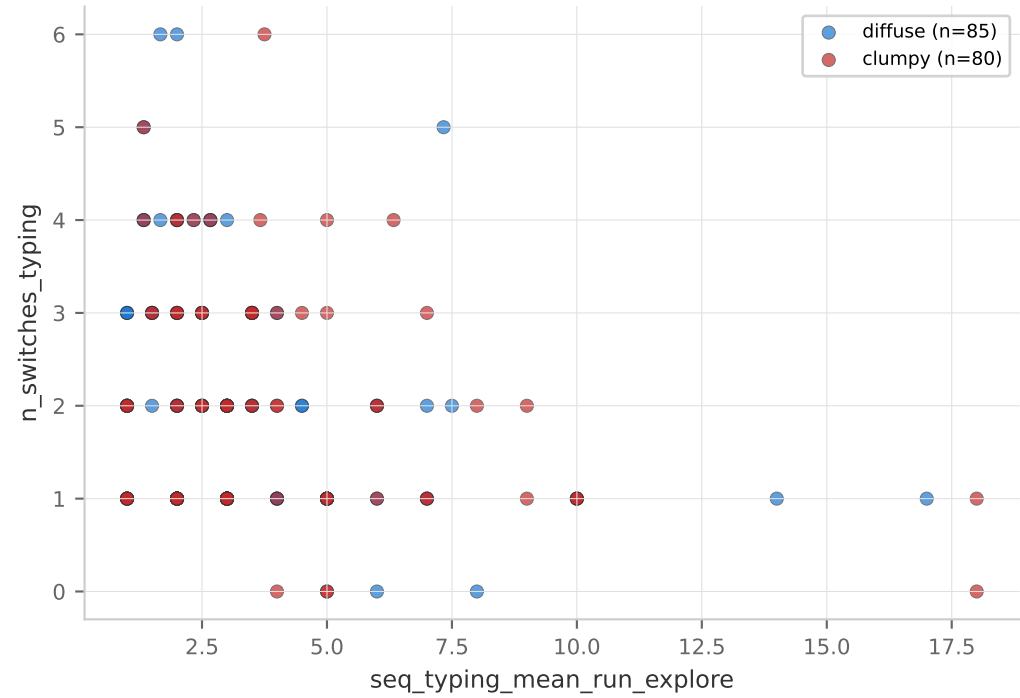
Switch rate = $n_switches / (n_pages - 1)$. Streak = mean run length of "no-typing" pages.
 If correlation is near -1 (high streak <-> low switches), the two measures are redundant.

M65 - Streak vs switch measures (scatter)

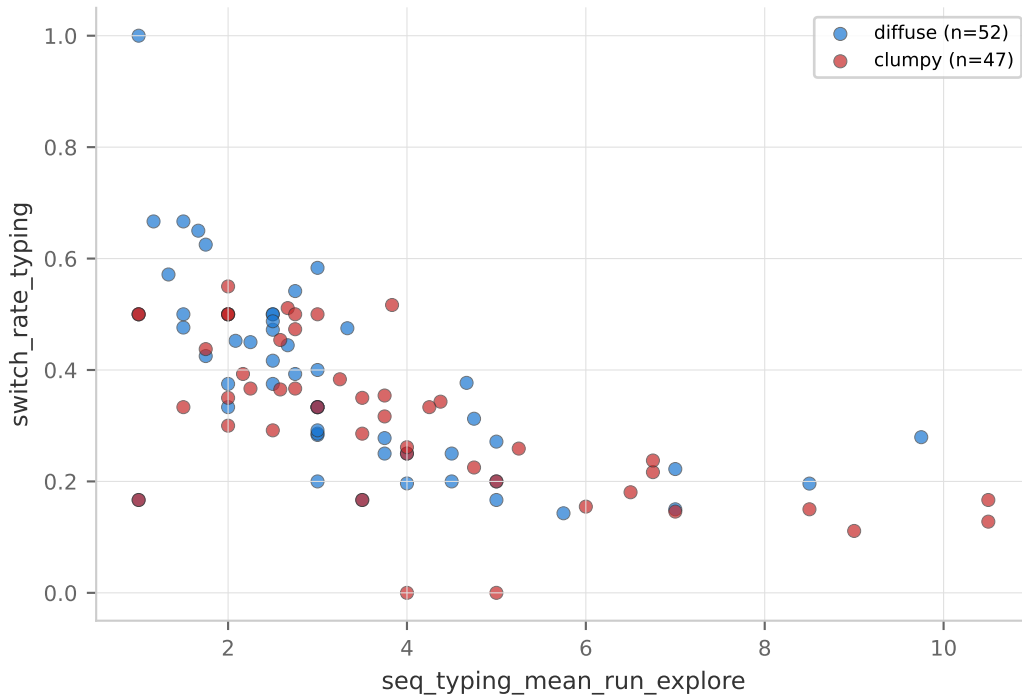
Per question (clean): streak vs switch rate
Pearson $r=-0.67$ $p=1.65e-22$



Per question (clean): streak vs n switches
Pearson $r=-0.29$ $p=0.000136$



Per participant (avg): streak vs switch rate
Pearson $r=-0.64$ $p=1.19e-12$



Per participant (avg): streak vs n switches
Pearson $r=-0.19$ $p=0.0585$

